

## Lab Procedure

# Passive Infrared

### Introduction

Ensure the following:

1. You have reviewed the **Application Guide – Passive Infrared**
2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.
  - a. The LED next to the USB C port will stay white after the touch screen starts loading.
  - b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

### Exploring Passive Infrared Sensors

1. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File > Open Folder...**, and browse to the directory containing the python files and lab procedure for the **Encoders—Passive Infrared** lab (`../sensors_trainer/1_fundamentals /passive_infrared/hardware/python`). Open this directory.

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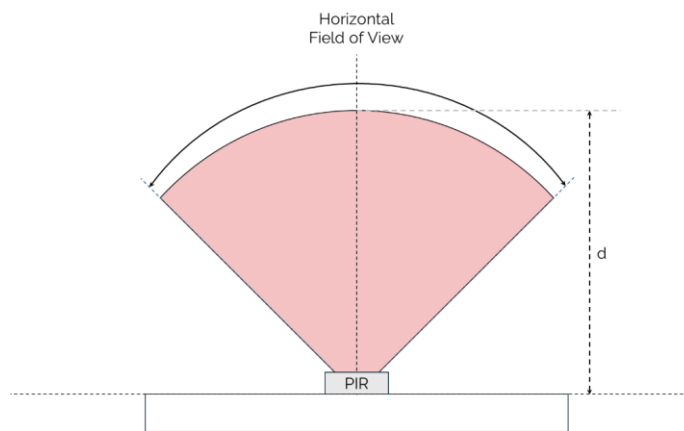
~~2. Open `passive_infrared.py`.~~

- ~~3. Print out a protractor on a 8.5' x 11' piece of paper. Within the (`../sensors_trainer/1_fundamentals /passive_infrared`) directory an example printable protractor is provided. Ensure away from the front surface of the **Mechatronics Sensors Trainer** and that the 0°-degree mark is aligned with the Passive Infrared sensor.~~

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4.3. For the first test we will estimate a field of view approximated as:



5. You may print/place a protractor such that the  $0^\circ$  is centered at the passive infrared location.

6. Make sure the mechatronics sensor trainer is connected via USB-C to a USB 3.0 port on a computer. Wait for the device to finish the start up sequence correctly.

7.4. Open the provided [Passive Infrared Datasheet](#) located in the passive infrared folder (`.../sensors_trainer/1_fundamentals/passive_infrared`) [PIR-PL-Q873-02.pdf](#) and double-check/verify the viewing angle for a passive infrared sensor with a natural lens cover.

8.1. Grab a warm (or cold) object such that there is a delta in temperature between ambient temperature and the object itself.

9.5. Open `passive_infrared.py`.

10.6. Click  to run the experiment.

11.7. Make a note of the voltage reading at ambient temperature.

12.8. Based on [trainer 018\\_passive\\_infrared](#) concept review, do you expect the field of view will be affected by small/large temperature changes? How do you expect the voltage reading will change if the delta of temperature between the object and the Passive Infrared sensor/ambient temperature is high? If it's low? Will the distance of the object also affect the field of view?

9. Grab a warm-(or-cold) object such that there is a delta in temperature between ambient temperature and the object itself.

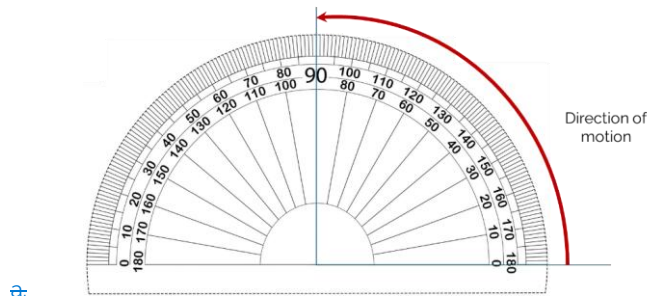
13.10. Re-run **passive\_infrared.py** and monitor the voltage reading.

**Commented [LG1]:** Is this just based on what they think? Something in the datasheet? Also does it detect cold objects? I have not tried it but I thought since its infrared it would only detect warmer things

**Commented [LG2R1]:** After reading the recommended assesment I realized the question is not if the voltage will increase or decrease. I don't fully know how to rephrase it, especially cause the datasheet doesn't even explain if there is more voltage at higher temperature change, just that its able to read it at larger distances, do we wanna ask something about that instead?

14.11. Being careful not to place your hand in the field of view near or in front of the Passive Infrared Sensor:

- a. Place the object at the 180° mark (on the right side of the protractor) on the protractor, roughly 0.5m away from the sensor board.



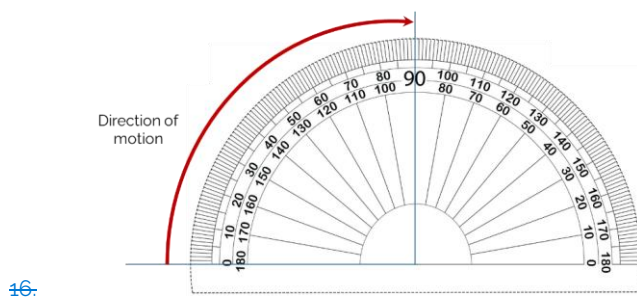
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- b. Slowly move the object closer towards to the 90° mark maintaining the 0.5m distance and while monitoring the voltage measurement from the Passive Infrared sensor.

15.12. When a voltage spike is measured/observed:

- a. stop moving the object,
- b. make a note of the angle at which the voltage spike occurred
- c. remove the object.

13. Redo steps 10 and 11 the previous 2 steps except this time place the object at the 0° mark on the left side and bring it closer towards to the 90° mark.



**Commented [LG3]:** Maybe I'd add one about what if you move the object from left to right quickly vs slowly, does it detect differently or does it detect no matter what.

I'd also try it with maybe like an arm or a torso that's covered since most of the time you are detecting a human moving

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17.14. What do you notice about the measured field of view compared to the field of view stated in PIR-PL-Q873-02.pdf shown in the datasheet?

18.15. Once the field of view extremes have been identified, place the object 1.5m away from the sensor at the 0° mark.

19.16. Re-run `passive_infrared.py` and monitor the voltage reading.

~~20~~17. Slowly move the object closer to the **Mechatronics Sensors Trainer**.

~~21~~18. When a voltage spike is measured:

- a. Stop moving the object,
- b. Make a note of distance and voltage values.

~~22~~19. Based on the tests performed, are Passive Infrared Sensors good for measuring distance, motion, both? Why?

~~23~~20. What are some system design considerations if you were to use a Passive Infrared sensor in a mechatronics application?

~~24~~21. Stop, save and close your program.

~~25~~22. Power OFF the Mechatronics Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.